# iGCSE Biology Study Guide: Characteristics & Variety of Living Organisms

## 📋 Course Overview & Mastery Tracking

Track your revision progress across all three levels. Check off each box once you have reviewed the content and successfully completed the practice questions.

### Level 1: The Basics

- [ ] Understand the 8 life processes shared by living organisms (**MRS C GREN**)
- [ ] Distinguish between **Eukaryotic** and **Prokaryotic** cellular structures
- [ ] Memorize the **5 Kingdoms** of living organisms and identify non-living entities (**Viruses**)

### Level 2: Deepening Understanding

- [ ] Compare structural features, nutrition modes, and carbohydrate storage across all 5 Kingdoms
- [ ] Detail the structure and lifecycle of **Fungi**, **Bacteria**, and **Viruses**
- [ ] Define **Pathogens** and identify disease-causing examples from each microscopic group

### Level 3: Expert Analysis & Exam Mastery

- [ ] Analyze edge cases (e.g., Viruses vs. Living Organisms, Photosynthetic Bacteria, Saprophytic Fungi)
- [ ] Complete exam-style scenario analyses and data interpretation questions
- [ ] Master multi-mark extended response strategies for classification and pathogens

# LEVEL 1: THE BASICS (FOUNDATION)

💡 **MIND-BLOWING FACT**

A giant fungal organism in Oregon's Malheur National Forest (*Armillaria ostoyae*) spans over 2,300 acres—the size of roughly 1,600 football fields! Despite its vast size, it is built from microscopic thread-like structures called hyphae, proving that all living things—no matter how giant or tiny—share fundamental biological rules.

## Section 1.1: The 8 Characteristics of Life (MRS C GREN)

Every living organism on Earth—from an Antarctic penguin to a deep-sea hydrothermal vent

bacterium—shares eight fundamental characteristics.

## The MRS C GREN Summary Flow

- **M**ovement → **R**espiration → **S**ensitivity → **C**ontrol → **G**rowth → **R**eproduction → **E**xcretion → **N**utrition

## Breakdown of the 8 Life Processes

1. **M — Movement**: An action by an organism or part of an organism causing a change of position or place.
   - *Animals*: Move their entire body to find food or escape predators.
   - *Plants*: Show slow directional growth movements (tropisms), such as turning leaves toward sunlight.
2. **R — Respiration**: A complex chemical reaction inside living cells that breaks down nutrient molecules (such as glucose) to release energy for cellular work.
   - *Equation*: $\text{Glucose} + \text{Oxygen} \rightarrow \text{Carbon Dioxide} +$
3. **S — Sensitivity**: The ability to detect or sense changes in the internal or external environment (stimuli) and to make appropriate responses.
   - *Example*: Detecting light, temperature, touch, or chemical changes.
4. **C — Control**: The ability to regulate and maintain stable internal conditions (homeostasis).
   - *Example*: Humans maintain a body temperature near $37^\circ\text{C}$ and regulate blood glucose levels. Plants regulate water loss through stomata.
5. **G — Growth**: A permanent increase in size and dry mass by an increase in cell number, cell size, or both.
6. **R — Reproduction**: The biological processes that produce new individuals of the same species (offspring), ensuring species survival.
   - Can occur via **sexual** (fusion of gametes) or **asexual** (cloning/binary fission) pathways.
7. **E — Excretion**: The removal from organisms of toxic materials, waste products of metabolism (chemical reactions in cells including respiration), and substances in excess of requirements.
   - *Key Examples*: Carbon dioxide, urea, water vapor.
8. **N — Nutrition**: Taking in materials for energy, growth, and development.
   - **Autotrophic**: Making own food (e.g., plants via photosynthesis).
   - **Heterotrophic**: Consuming other living organisms (e.g., animals eating plants or meat).

### ⚠️ EXAM TRAP ALERT

Do **NOT** confuse **Excretion** with **Egestion**!

- **Excretion** is the removal of metabolic waste produced *inside cells* (e.g., $CO_2$ from respiration, urea from protein breakdown).
- **Egestion** is the removal of undigested food as feces through the anus—this material never

entered body cells!

# Section 1.2: Eukaryotes vs. Prokaryotes

All cellular organisms fall into two main structural categories based on whether their genetic material is enclosed within a nucleus.

| Structural Feature | Eukaryotic Organisms | Prokaryotic Organisms |
|---|---|---|
| **Nucleus** | **Present** (DNA enclosed in nuclear membrane) | **Absent** (DNA floats freely in cytoplasm) |
| **Genetic Material** | Linear chromosomes bound to proteins | Single circular chromosome + small rings of DNA (**plasmids**) |
| **Membrane-bound Organelles** | Present (mitochondria, chloroplasts, endoplasmic reticulum) | Absent |
| **Cell Diameter** | Larger ($10–100\ \mu m$) | Smaller ($0.5–5\ \mu m$) |
| **Cell Wall Structure** | Cellulose (plants), Chitin (fungi), None (animals) | **Peptidoglycan** |
| **Kingdom Representation** | Animals, Plants, Fungi, Protoctists | Bacteria |

🔑 **MEMORY ANCHOR**

- **EU**karyote = **EU** (You!) have a nucleus.
- **PRO**karyote = **PRO** (Before) a nucleus existed.

# 1.3 Cover & Test Yourself (Level 1)

1. State the 8 life processes represented by the mnemonic **MRS C GREN**.
2. Which organelle is present in eukaryotic cells but completely absent in prokaryotic cells?
3. Is a plant cell classified as eukaryotic or prokaryotic? Explain why.

**Answers**:

1. **M**ovement, **R**espiration, **S**ensitivity, **C**ontrol, **G**rowth, **R**eproduction, **E**xcretion, **N**utrition.
2. The **nucleus** (or membrane-bound organelles such as mitochondria/chloroplasts).
3. **Eukaryotic**, because its genetic material is enclosed within a membrane-bound nucleus and it possesses membrane-bound organelles.

# LEVEL 2: DEEPENING UNDERSTANDING (INTERMEDIATE)

💡 **REAL-WORLD CONNECTION**

In Sub-Saharan Africa and Southeast Asia, farmers rely on legumes (pea and bean crops) to naturally enrich soil nitrogen without synthetic fertilizers. The root nodules of these plants host nitrogen-fixing bacteria (*Rhizobium*). This symbiotic relationship between a **Plant** and a **Bacteria** showcases inter-kingdom interaction.

## Section 2.1: The 5 Kingdoms Breakdown

Living organisms are classified into 5 distinct Kingdoms. Non-living biological agents like **Viruses** are excluded from this system.

### Classification Tree Overview

- **All Living Organisms**
  - **Eukaryotic Group** (Possesses Nucleus)
    - Plants
    - Animals
    - Fungi
    - Protoctists
  - **Prokaryotic Group** (Lacks Nucleus)
    - Bacteria

### Comparative Kingdom Matrix

| Kingdom | Cellularity | Cell Wall Composition | Photosynthetic? | Nutrition Type | Carbohydrate Storage | Key Examples |
|---|---|---|---|---|---|---|
| **Plants** | Multicellular | **Cellulose** | Yes (contain chloroplasts) | Autotrophic | **Starch or Sucrose** | Maize, Pea, Mosses, Ferns |
| **Animals** | Multicellular | **None** | No | Heterotrophic | **Glycogen** | Humans, Insects, Jellyfish, Coral |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Fungi** | Multicellular or Unicellular | **Chitin** | No | Saprophytic (extracellular digestion) | **Glycogen** | Yeast (unicellular), Mucor, Mushrooms |
| **Protoctists** | Mostly Unicellular | Varies (cellulose in algae) | Some (*Chlorella*) | Autotrophic or Heterotrophic | Varies | *Amoeba* (animal-like), *Chlorella* (plant-like), *Plasmodium* |
| **Bacteria** | Unicellular | **Peptidoglycan** | Some species | Autotrophic or Saprophytic | Varies | *Lactobacillus bulgaricus*, *Pneumococcus* |

# Section 2.2: Deep Dive into Each Kingdom

## 1. Kingdom Plants

- **Key Characteristics**: Multicellular organisms that carry out photosynthesis to synthesize organic nutrients ( $C_6H_{12}O_6$ ).
- **Storage**: Store carbohydrates as **starch** or **sucrose**.
- **Major Groups**:
  - *Flowering Plants*: Maize (monocotyledon), Pea (dicotyledon).
  - *Non-flowering Plants*: Mosses, Ferns.

## 2. Kingdom Animals

- **Key Characteristics**: Multicellular organisms; cells lack cell walls and chloroplasts.
- **Coordination & Locomotion**: Possess **nervous coordination** and are capable of movement from one place to another.
- **Storage**: Store carbohydrates as **glycogen**.
- **Major Groups**:
  - *Vertebrates* (with backbone): Mammals (e.g., Polar bears), Birds, Reptiles, Amphibians, Fish.

    - *Invertebrates* (without backbone): Insects (e.g., Mosquitoes), Molluscs (e.g., Snails), Annelids (e.g., Earthworms), Crustaceans.

## 3. Kingdom Fungi

- **Key Characteristics**: Can be multicellular (e.g., *Mucor*, Mushrooms) or unicellular (e.g., *Yeast*). Cells never contain chloroplasts.
- **Structure**: Body of multicellular fungi is organized into a network of thread-like structures called **hyphae**, which contain many nuclei; the entire network is called a **mycelium**.
- **Cell Wall**: Constructed of **chitin**.
- **Saprophytic Nutrition**: Fungi feed by secreting **extracellular digestive enzymes** onto dead or decaying organic material, breaking it down outside the body, and then absorbing the soluble digested products.
- **Storage**: Store carbohydrates as **glycogen**.

## Saprophytic Digestion Process

1. **Enzyme Secretion**: Hyphae release digestive enzymes onto dead substrate.
2. **Extracellular Breakdown**: Enzymes digest complex organic matter externally into simple soluble molecules.
3. **Absorption**: Fungal cells absorb the soluble nutrients across chitin cell walls into cytoplasm.

**🔑 EXAM CHEAT CODE**

When describing fungal nutrition on an exam, use these exact keywords in order:

1. **Secretes extracellular digestive enzymes**
2. **Onto dead organic matter**
3. **Absorbs soluble products**

## 4. Kingdom Protoctists

- **Key Characteristics**: A collection of simple eukaryotic organisms that do not fit neatly into the plant, animal, or fungal kingdoms. Known as the "dustbin kingdom."
- **Cellular Structure**: Mostly single-celled (unicellular), though some seaweeds are multicellular.
- **Varieties**:
    - *Animal-like Protoctists*: Live in water, feed on other organisms (e.g., *Amoeba*).
    - *Plant-like Protoctists*: Contain chloroplasts and photosynthesize (e.g., *Chlorella*).
    - *Pathogenic Protoctists*: Microscopic parasites causing disease (e.g., *Plasmodium*, which causes malaria).

## 5. Kingdom Bacteria (Prokaryotes)

- **Key Characteristics**: Microscopic single-celled organisms lacking a true membrane-bound nucleus and organelles.

- **Internal Features**:
  - **Cell Wall**: Made of **peptidoglycan**.
  - **Cell Membrane & Cytoplasm**: Present.
  - **Genetic Material**: No nucleus; possesses a **single circular chromosome** of DNA located free in the cytoplasm.
  - **Plasmids**: Small, circular loops of extra DNA that can carry beneficial genes (e.g., antibiotic resistance).
  - **Structures for Movement**: Some have **flagella**.
- **Nutrition**: Most feed on other living or dead organisms (saprophytes/heterotrophs); some contain chlorophyll and can perform photosynthesis.
- **Key Examples**:
  - *Lactobacillus bulgaricus*: Rod-shaped bacterium used in yoghurt production from milk.
  - *Pneumococcus*: Spherical bacterium that acts as a pathogen causing pneumonia.

# Section 2.3: Non-Living Entities — Viruses

Viruses are **NOT** included in the five kingdoms because they are **non-living biological structures**.

## Infection Cycle Overview

- Virus Particle → Binds to Host Cell Surface → Injects DNA/RNA → Host Machinery Replicates Viruses

## Why Are Viruses Classified as Non-Living?

1. **No Cellular Structure**: They are particles, not cells (no cytoplasm, cell membrane, or organelles).
2. **Do Not Perform Life Processes**: They do not move, respire, excrete, grow, respond to stimuli, or feed on their own.
3. **Parasitic Reproduction**: They can **only reproduce inside living host cells**, taking over the host cell's genetic machinery to replicate.

## Key Features of Viruses

- **Size**: Ultra-microscopic (much smaller than bacteria, typically $20–300\,\text{nm}$ ).
- **Structure**: A core of genetic material (**either DNA or RNA**, never both) surrounded by a **protein coat (capsid)**.
- **Hosts**: They infect **every type of living organism** (animals, plants, fungi, protoctists, and bacteria).

## Key Examples of Viruses

- *Tobacco Mosaic Virus (TMV)*: Prevents chloroplast formation in tobacco leaves, causing mottling/discoloration and reducing growth.
- *Influenza Virus*: Causes the flu in mammals.

- *HIV (Human Immunodeficiency Virus)*: Attacks immune cells, leading to AIDS.
- *SARS-CoV-2*: Causes COVID-19.

# Section 2.4: Pathogens Overview

A **pathogen** is defined as an organism or agent that causes disease in its host. Pathogens span four biological categories:

1. **Pathogenic Fungi**: e.g., *Tinea pedis* (causes Athlete's foot).
2. **Pathogenic Bacteria**: e.g., *Pneumococcus* (pneumonia), *Vibrio cholerae* (cholera).
3. **Pathogenic Protoctists**: e.g., *Plasmodium* (causes malaria, transmitted by mosquito vectors).
4. **Pathogenic Viruses**: **ALL natural viruses are pathogens** because they cause harm to their host cells during replication.

# 2.5 Cover & Test Yourself (Level 2)

1. Complete the missing cells in the comparison table below:

| **Group** | **Cell Wall Material** | **Carbohydrate Storage Form** |
|---|---|---|
| **Plants** | (a) _____ | Starch / Sucrose |
| **Animals** | None | (b) _____ |
| **Fungi** | (c) _____ | Glycogen |
| **Bacteria** | (d) _____ | Varies |

2. Describe how a fungus obtains food via saprophytic nutrition.
3. State three structural differences between a bacterium and a virus.

**Answers**:

1. (a) **Cellulose**, (b) **Glycogen**, (c) **Chitin**, (d) **Peptidoglycan**.
2. A fungus **secretes extracellular digestive enzymes** onto dead organic material. These enzymes digest the substrate outside the organism, and the fungus then **absorbs the soluble products**.
3. Any three from:
    - Bacteria have a cellular structure (cytoplasm/membrane); viruses are non-cellular particles.
    - Bacteria have cell walls made of peptidoglycan; viruses have a protein coat (capsid).
    - Bacteria contain both DNA and RNA; viruses contain either DNA or RNA (never both).
    - Bacteria possess plasmids and ribosomes; viruses lack all cell organelles.
    - Bacteria are larger than viruses.

# LEVEL 3: EXPERT ANALYSIS & EXAM MASTERY

💡 **REAL-WORLD CONNECTION**

*Amoeba proteus* is a single-celled protoctist found in freshwater ponds around the globe—from the Nile River to the Amazon basin. To survive without bursting, it continuously collects excess water entering its cell via osmosis and pumps it out using a specialized structure called a **contractile vacuole**. This is a real-world demonstration of **Control** (homeostasis) in a single eukaryotic cell!

## Section 3.1: Complex Applied Concepts

On iGCSE exams, extended response questions often ask you to evaluate whether viruses are "living" organisms. Use the structured decision framework below to earn maximum marks:

🎯 **EXAM CHEAT CODE: The 5-Mark Virus Response Strategy**

When asked why a virus is excluded from the 5 Kingdoms, structure your answer into two clear sections:

**Part A: Absence of Life Characteristics (3 marks)**

- Mention at least 3 life processes the virus **cannot** do on its own (e.g., does not respire, excrete, feed, grow, or respond to stimuli).
- State that it can **only reproduce inside living host cells**.

**Part B: Absence of Cellular Structures (2 marks)**

- State that it lacks cytoplasm, cell membrane, and organelles.
- State that it consists only of a **protein coat** and **one type of nucleic acid** (DNA or RNA).

## Section 3.2: Comprehensive Summary Table for Quick Revision

| Kingdom | Nucleus Present? | Cell Wall Type | Photosynthetic? | Primary Carbohydrate Storage |
|---|---|---|---|---|
| **Plants** | Yes (Eukaryotic) | Cellulose | Yes | Starch / Sucrose |
| **Animals** | Yes | None | No | Glycogen |

| | | | | |
|---|---|---|---|---|
| | (Eukaryotic) | | | |
| **Fungi** | Yes (Eukaryotic) | Chitin | No | Glycogen |
| **Protoctists** | Yes (Eukaryotic) | Varies | Some species | Varies |
| **Bacteria** | No (Prokaryotic) | Peptidoglycan | Some species | Varies |

# Section 3.3: Exam-Style Question Set

## Question 1: Comparative Data Analysis [5 Marks]

Complete the missing entries (A–E) in the following table:

| Kingdom | Carbohydrate Storage Form | Cell Wall Composition | Nucleus Present? |
|---|---|---|---|
| **Animal** | Glycogen | **A** | Yes |
| **Plant** | **B** | Cellulose | Yes |
| **Bacteria** | Varies | **C** | **D** |
| **Fungi** | Glycogen | **E** | Yes |

- **A**: None [1 mark]
- **B**: Starch (or Sucrose) [1 mark]
- **C**: Peptidoglycan [1 mark]
- **D**: No [1 mark]
- **E**: Chitin [1 mark]

## Question 2: Passage-Based Comprehension — Jellyfish Biology [5 Marks]

*Read the following passage carefully:*

"Jellyfish are aquatic organisms found across world oceans. They swim using rhythmic pulsations of their umbrella-shaped body. Tentacles lined with millions of stinging cells trap small fish and zooplankton for food. Jellyfish take in oxygen directly across their thin tissue layers from surrounding water and release carbon dioxide back into the sea. Lacking a central brain, they coordinate movement using a diffuse nerve net. Most reproduce sexually by releasing eggs

and sperm, but some species can clone themselves asexually."

**Q2a**: Suggest and explain **three** pieces of evidence from the passage showing that jellyfish are living organisms. [3 marks]

Award 1 mark for each valid trait paired with correct evidence from text (Max 3):

1. **Movement**: Rhythmic pulsations of the umbrella body to swim.
2. **Nutrition**: Trapping small fish and zooplankton using stinging tentacles to feed.
3. **Respiration**: Taking in oxygen across tissues for cellular respiration.
4. **Sensitivity / Control**: Using a nerve net to coordinate responses and movement.
5. **Reproduction**: Releasing eggs and sperm for sexual reproduction (or cloning asexually).

**Q2b**: Name the process by which carbon dioxide is lost from jellyfish cells into the water, and state which life process this represents. [2 marks]

- **Process**: Diffusion [1 mark]
- **Life Process**: Excretion [1 mark]

## Question 3: Extended Written Answer [5 Marks]

**Explain why viruses are not classified in any of the five kingdoms of living organisms.**

*Any 5 points from the following list (Max 5):*

1. Viruses do not carry out metabolic life processes independently (e.g., respiration, excretion, nutrition, growth, sensitivity, movement) [1 mark]
2. They can **only reproduce inside living host cells** (parasitic) [1 mark]
3. They lack a cellular structure / are acellular [1 mark]
4. They do not have cytoplasm or a cell membrane [1 mark]
5. They do not have cell organelles (e.g., mitochondria, ribosomes) [1 mark]
6. They consist only of a genetic core (DNA or RNA) surrounded by a protein coat (capsid) [1 mark]

# 🌐 RECOMMENDED OPEN REVISION RESOURCES

If you want to know more about this, visit these sites:

1. **PhET Interactive Simulations (University of Colorado Boulder)** *Interactive virtual lab simulations for cell structure and microscopic dynamics.* https://phet.colorado.edu
2. **BBC Bitesize — GCSE Biology: Characteristics of Living Organisms** *Comprehensive revision notes, clear visual diagrams, and quick self-assessment quizzes for iGCSE biology.* https://www.bbc.co.uk/bitesize/subjects/z9ddmp3
3. **Khan Academy — Biology Library: Microorganisms and Classification** *In-depth video tutorials and structured articles explaining eukaryotic vs. prokaryotic structures and*

*pathogens.* https://www.khanacademy.org/science/biology

4. **CK-12 Foundation — Life Science for Middle & High School** *Free open-access textbook chapters and interactive modalities covering the 5 kingdoms and viral structures.* https://www.ck12.org/biology/
5. **Microbiology Society — Education Resources** *Free downloadable guides and educational articles on bacteria, fungi, viruses, and pathogens.* https://microbiologysociety.org
6. **Biology LibreTexts — General Biology & Microbiology** *Comprehensive, peer-reviewed open textbook modules for deep dives into saprophytic fungal nutrition and prokaryotic cell structures.* https://bio.libretexts.org